

Sampling

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LEARNING OBJECTIVES

- 1. Define sampling, sample, population, element, and subject**
- 2. The sampling process**
- 3. Probability sampling & Nonprobability sampling designs**

INTRODUCTION



INTRODUCTION

- => The process of selecting the right individuals, objects, or events as representatives for the entire population is known as **SAMPLING**
 - ✓ Select SAMPLES for POPULATION
 - ✓ Purpose: use data of samples to make conclusion/ estimation of the POPULATION
 - ✓ Advantage: Save the time, cost, and other human resources

Main terms

- In learning how representative data can be collected, these terms must first to be understood:
 - **POPULATION**
 - **ELEMENT**
 - **SAMPLE**
 - **SUBJECT**

Population

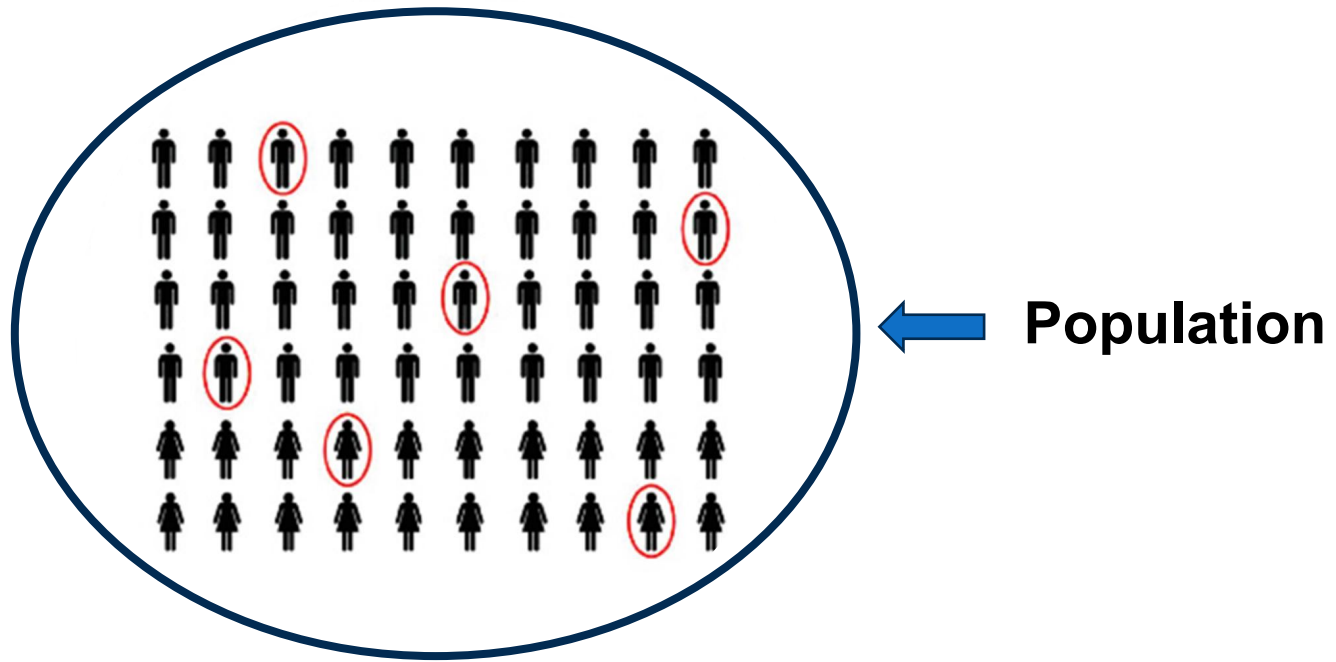


- Students in a University A
- People in a City B
- People in the whole world
- ...



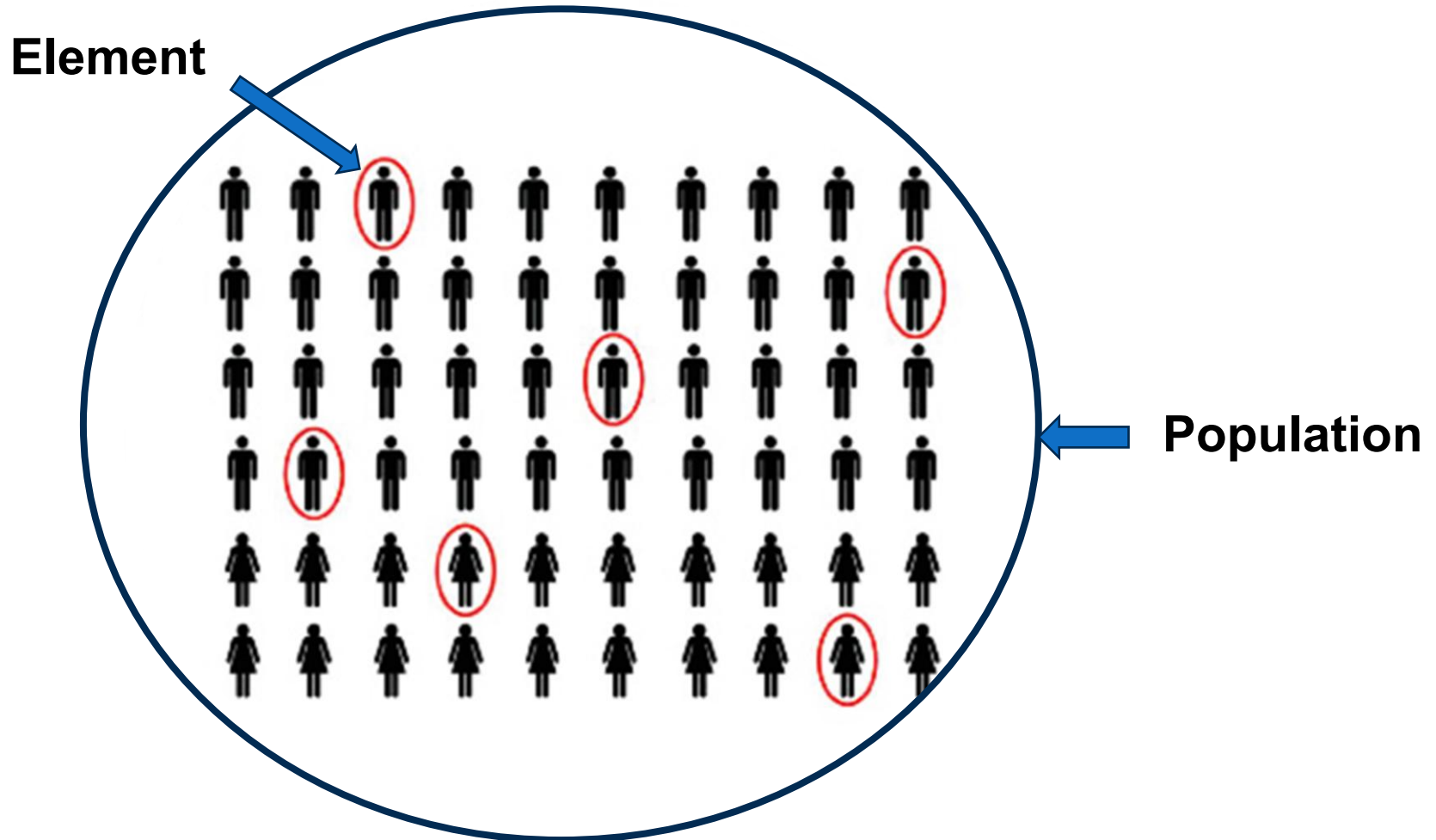
Population

- The population refers to the entire group of people, events, or things of interest that the **researcher wishes to investigate and make inferences**.



Element

- An element is a single member of the population



Sample

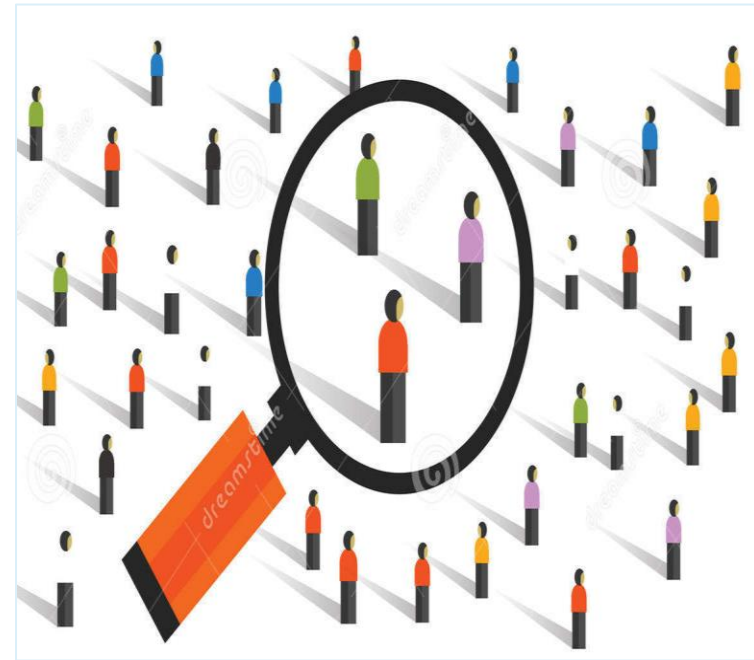
Target Population



Sample

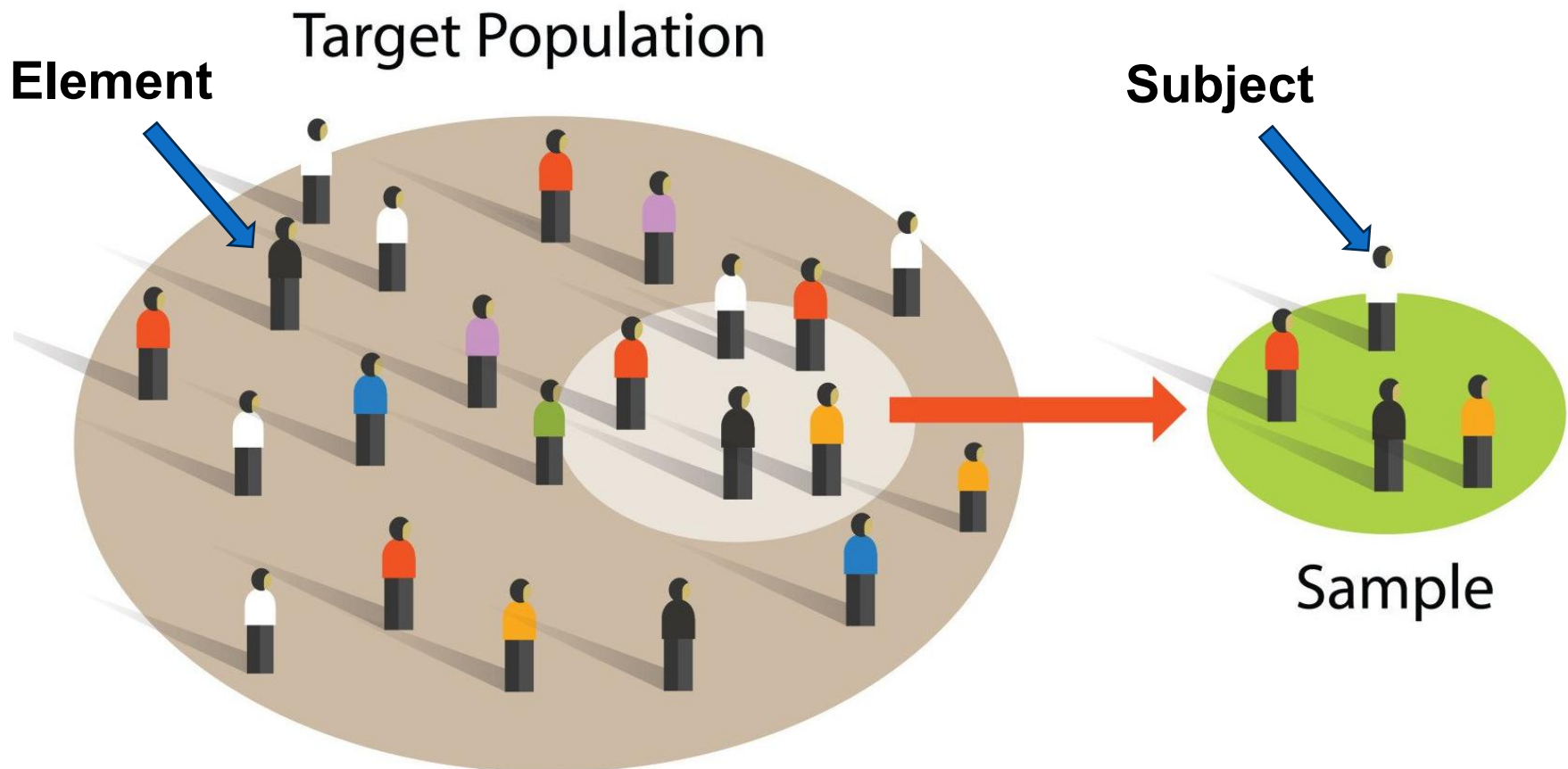
Sample

- A sample is a subset of the population. It comprises some members selected from it
 - ✓ In other words, elements of the population form the sample
- By studying the sample, the researcher should be able to draw conclusions that are generalizable to the population of interest

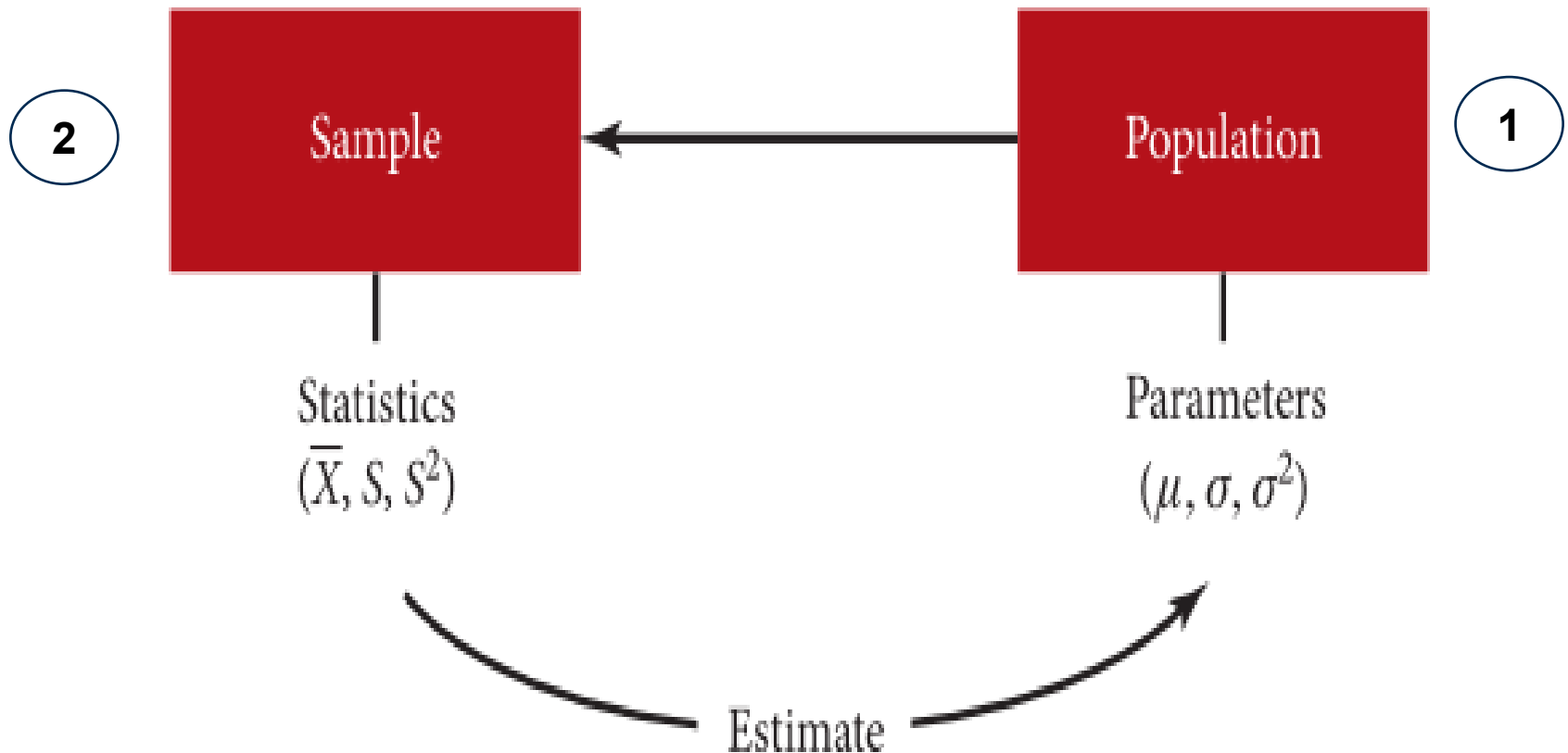


Subject

- **A subject is a single member of the sample**
 - ✓ just as an element is a single member of the population



Representativeness of Samples



THE SAMPLING PROCESS

- **Sampling is the process of selecting a sufficient number of the right elements from the population**
- **The major steps in sampling include:**
 - ✓ 1. Define the POPULATION
 - ✓ 2. Determine the SAMPLE FRAME
 - ✓ 3. Determine the appropriate SAMPLE SIZE
 - ✓ 4. Determine the SAMPLING DESIGN
 - ✓ 5. Execute the SAMPLING PROCESS

Defining the POPULATION

- **Sampling begins with defining the target population**
- **The target population must be defined in terms: elements, geographical boundaries, and time**
- **Example:**
 - ✓ For a banker interested in saving habits of blue-collar workers in the mining industry in the United States, the target population might be all blue-collar workers in that industry throughout the country



Determining the SAMPLE FRAME

- The sampling frame is a physical or digital list of all the elements in the population from which the sample is drawn.
- Using the sample frame for **Probability sampling**



Determining the SAMPLE SIZE

Sample size know about population: (Slovin)

$$n = \frac{N}{1 + N \times e^2}$$

- N : number of population.
- e : acceptance error

Determining the SAMPLE SIZE

Sample size know about population: (Slovin)

Exercise: A researcher wants to study the job satisfaction of employees at a local manufacturing company. The company has a total of **1,000 employees**. Due to time and budget constraints, the researcher cannot interview everyone and needs to determine the appropriate **sample size (n)**. Please calculate **the sample size** of this research. Acceptant error is 5%.

$$n = \frac{N}{1 + N \times e^2}$$

- N : number of population.
- e : acceptance error

Determining the SAMPLE SIZE

Sample size don't know about population: (Cochran)

SAMPLE SIZE FORMULA COMPONENTS

$$\text{Sample Size} = \frac{z^2 \times p(1-p)}{e^2}$$

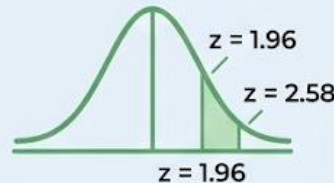
CALCULATE IDEAL SAMPLE SIZE

MARGIN OF
ERROR (e)



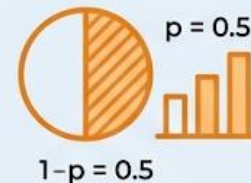
SAI SỐ CHO PHÉP (e)

Z SCORE (z) &
CONFIDENCE LEVEL



GIÁ TRỊ TỐI HẠN Z (z)

ESTIMATED
PROPORTION (p)



TỶ LỆ DỰ KIẾN (p)

e = Margin of error

z = Z score

p = Estimated proportion (often assumed 0.5)



Determining the SAMPLE SIZE

Sample size don't know about population: (Cochran)

Exercise: A researcher wants to conduct a national survey to find out the percentage of digital nomads who prefer working from coastal cities. Since the total number of digital nomads worldwide is **unknown or extremely large (infinite)**, the researcher uses Cochran's formula to determine the required sample size (n).

$$\text{Sample Size} = \frac{z^2 \times p(1 - p)}{e^2}$$

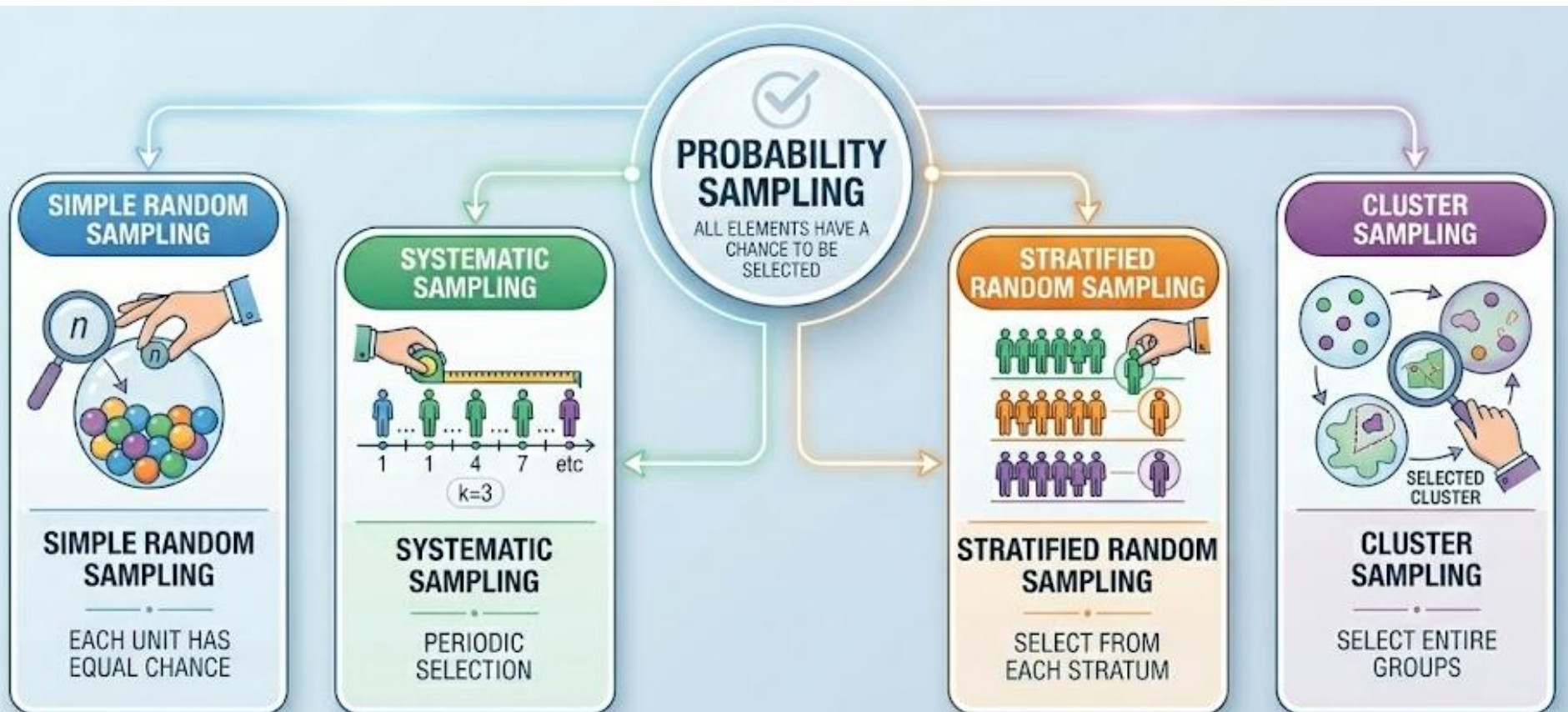
Percentage Confidence	z*-Value
80	1.28
90	1.645
95	1.96
98	2.33
99	2.58

Determining the SAMPLING DESIGN

- The choice of the sampling procedure is a very important
- There are two major types of sampling design:
 - ✓ *Probability sampling*: the elements in the population have some known or probability of being selected as sample subjects
 - ✓ *Non-probability sampling*: the elements do not have a known or predetermined chance of being selected as subjects

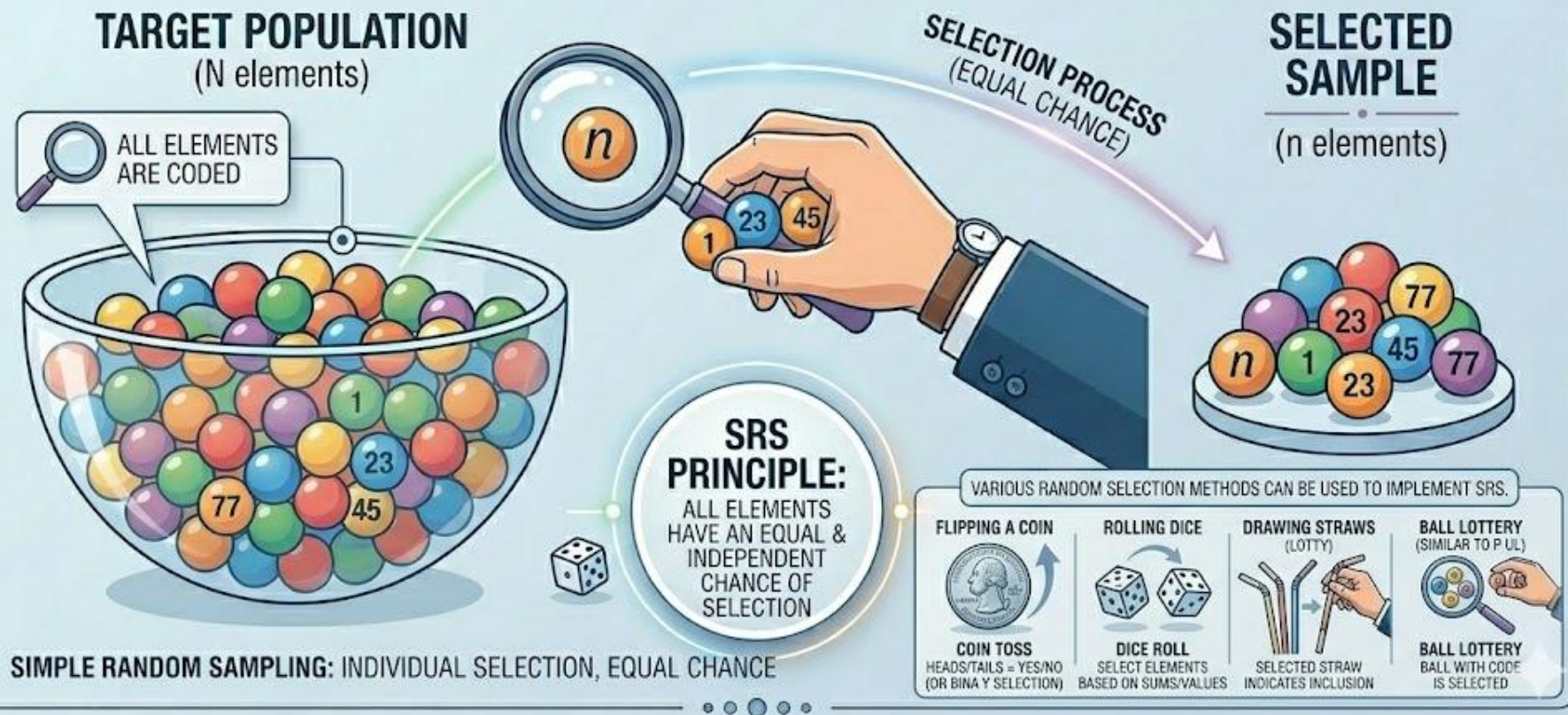
(1) Probability sampling

- When elements in the population have a known, nonzero chance of being chosen as subjects in the sample, we resort to a probability sampling design



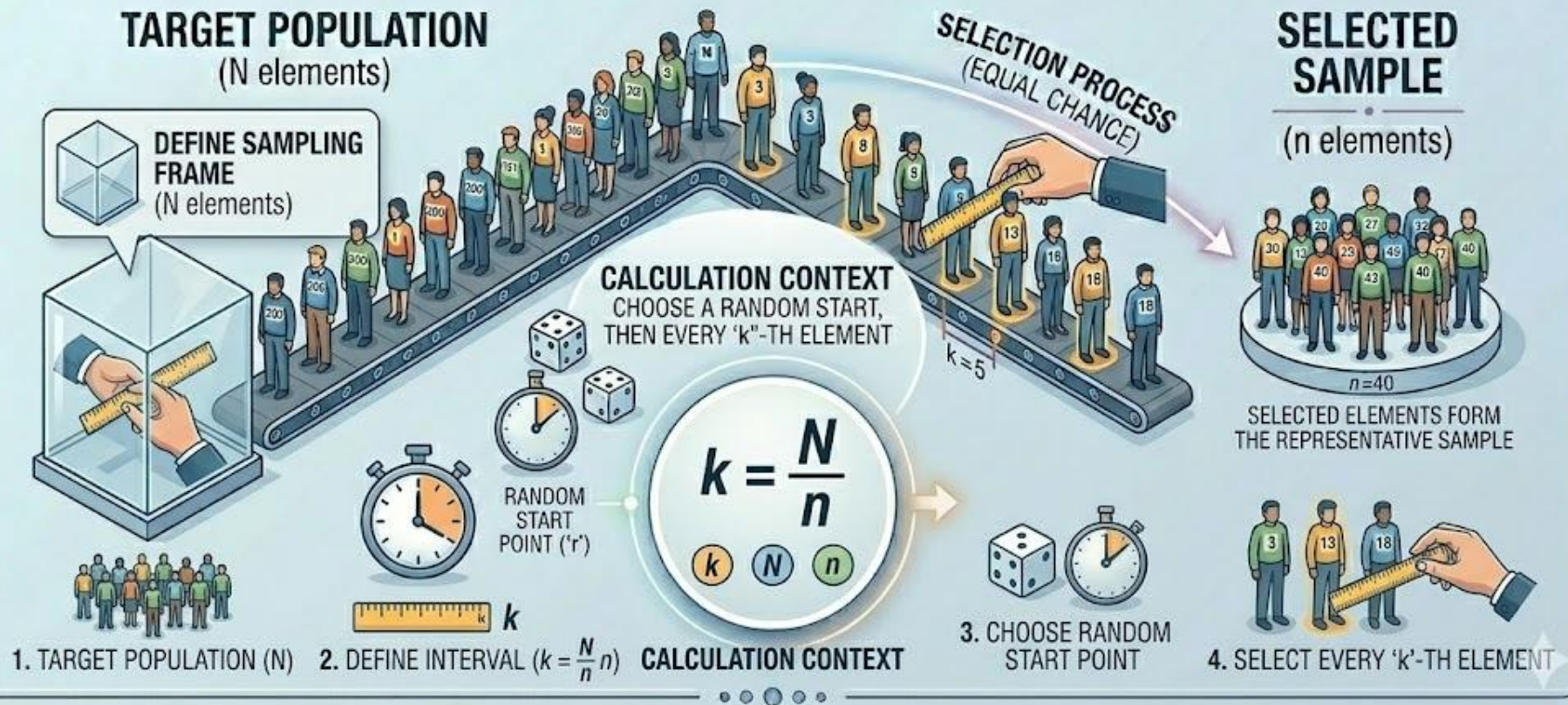
(1) Probability sampling

.. SIMPLE RANDOM SAMPLING (SRS) ..



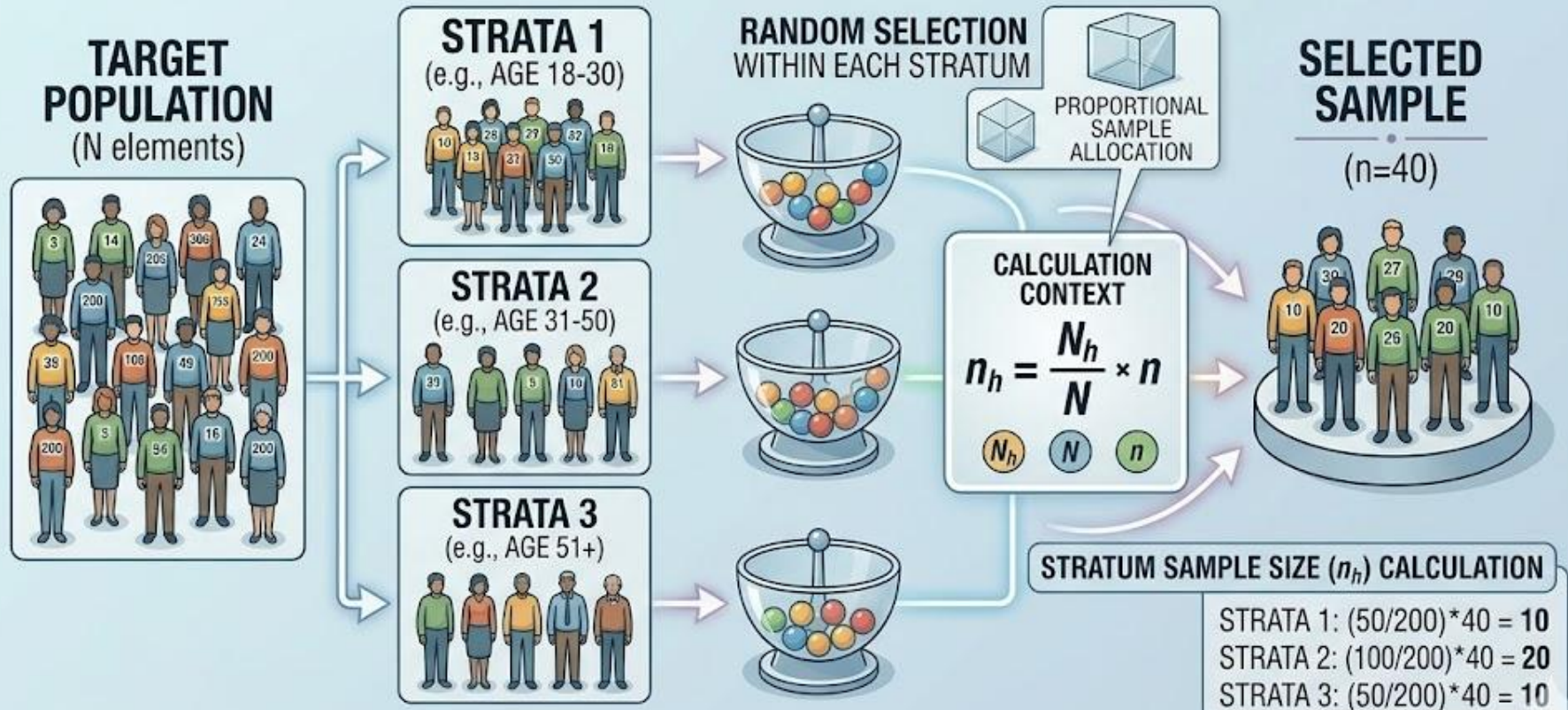
(1) Probability sampling

SYSTEMATIC SAMPLING (SS)



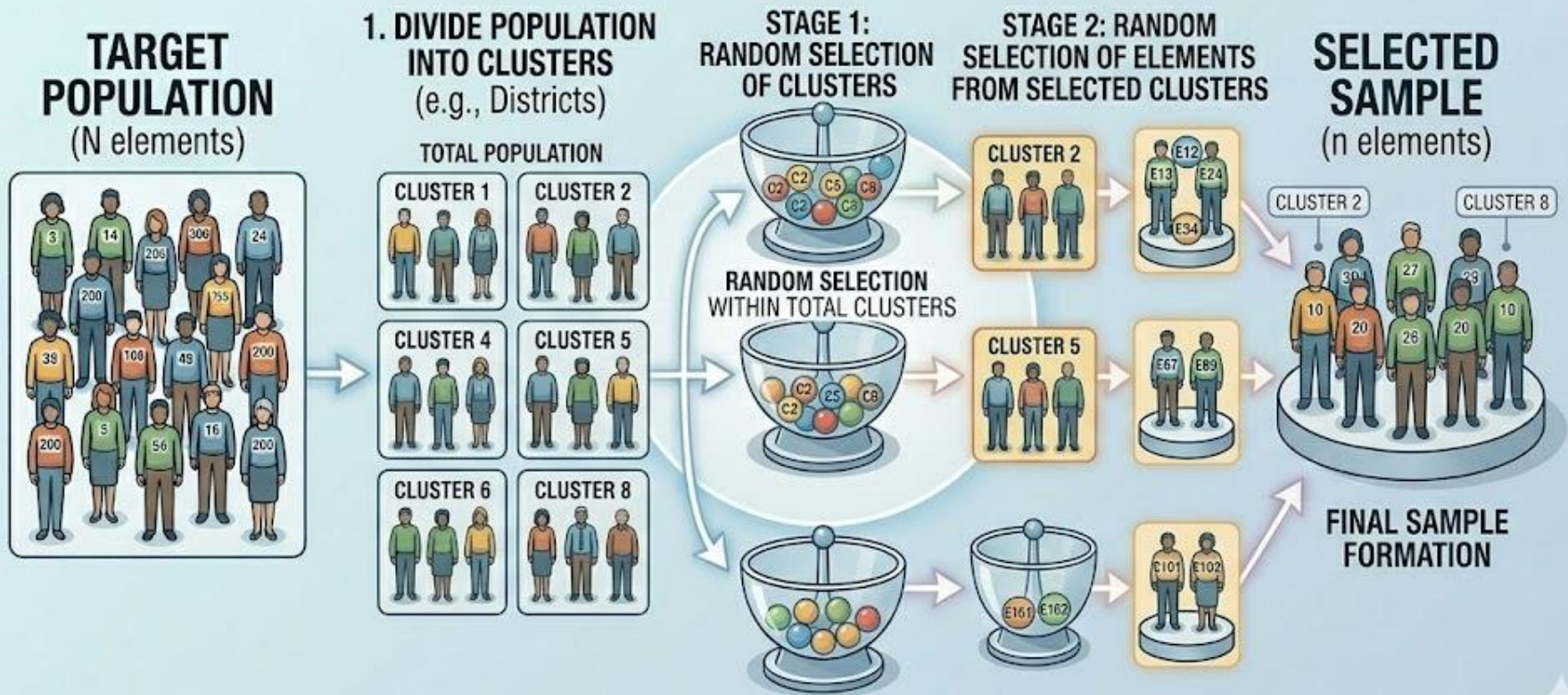
(1) Probability sampling

STRATIFIED SAMPLING METHOD



(1) Probability sampling

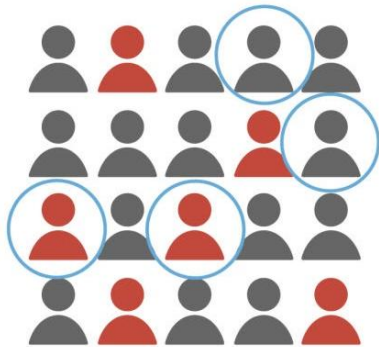
CLUSTER SAMPLING METHOD



(2) Nonprobability sampling

- **Nonprobability sampling designs: the elements in the same population have zero chance of being chosen as subjects**
 - ✓ This means that the findings from the study of the sample cannot be confidently generalized to the population
 - ✓ There are two types:
 - Convenience sampling
 - Purposive sampling

(2) Nonprobability sampling



Convenience Sample



Purposive Sample

Nonprobability sampling (cont)

• Convenience sampling:

• 1. RECRUITING FRIENDS & FAMILY •



• 2. INTERCEPTING IN PUBLIC SPACES •



How to
Select
Sample

• 3. ASKING FOR VOLUNTEERS •



• 4. RECRUITING ONLINE / VIA SOCIAL MEDIA •



Nonprobability sampling (cont)

- **Purposive sampling:**

- ✓ Obtain information from specific target groups
- ✓ The sampling here is confined to specific types of people who can provide the desired information

Exercise

Please choose **the sampling design** and **the sample size** of your research topic.
And describe the sampling process.

**THANK YOU
FOR THE KIND ATTENTION!**